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REVIEW ARTICLE

Energy production from farming waste: a review

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Abstract
The global increase in greenhouse gas emissions is mainly due to electricity and heat generation, transportation, manufacturing and construction, and agriculture. In particular, agriculture produces underutilized waste that generates greenhouse gas emissions and other pollutants. Here we review the conversion of farming waste into energy, with focus on waste from agriculture, livestock, poultry, and pisciculture. Energy production techniques include pelletization, pyrolysis, gasification, liquefaction, anaerobic digestion, and fermentation. In thermochemical processes, the biofuel yield is controlled by temperature, pressure, heating rate, feedstock concentration, reaction time, catalysts, and reactor type. In biological processes, the biofuel yield is controlled by pretreatment intensity, microorganisms, substrate loading, temperature, volatile solids, reaction time, and inhibitors.

Keywords Anaerobic digestion · Fermentation · Gasification · Liquefaction · Pelletization · Pyrolysis

Introduction

The current global population of 8.2 billion people is expected to rise to 9 billion by 2037 and 10 billion by 2060 [1]. As the world's population grows, the demand for necessities such as food, potable water, clean air, shelter, and energy sources like fuel and electricity also tends to increase. The agricultural sector, the backbone of our civilization, plays a pivotal role in meeting the world's food demands. However, this sector also generates a substantial amount of crop residues such as stalks, husk, straw, and other agricultural byproducts, which are often left unutilized, discarded, or burnt after harvest, releasing significant greenhouse gas emissions. In addition, poultry, livestock, and fish culture are growing simultaneously to meet our protein demand.

Various farming practices, such as agriculture, poultry, livestock, and fishery, produce vast amounts of waste, including manure, bedding materials, and slaughterhouse waste, as illustrated in Figure 1. The accumulation and mismanagement of these residues often lead to environmental issues like air pollution, soil degradation, greenhouse gas emissions, odor generation, and the occurrence of pests, rodents, and insects. However, the innovative concept of harnessing this farm residue for bioenergy production offers a sustainable solution to this challenge. Approximately 53 billion metric tons of CO_2 equivalent is emitted globally compared to 24 billion metric tons in 1970 [2]. Figure 2 shows the trend of per capita greenhouse gas emissions from different sectors on a global scale. The electricity and heat generation sector contributes to 2 tons of CO_2 equivalent per capita, followed by the transportation sector, attributing to 1 ton of CO_2 equivalent per capita [3]. Agriculture contributes to 0.7 ton of CO_2 equivalent per capita emission.

The global energy demand and consumption are rising with industrialization, urbanization, and rapid population growth. Moreover, several environmental challenges such as climate change, ozone layer depletion, and greenhouse gas emissions have led to the focus on renewable sources of energy [4]. Renewable resources like wind, solar, and hydro energy are alternative substitutes for fossil fuels. Still, they cannot generate liquid or gaseous fuels for vehicles or combined heat and power plants. Additionally, the renewable energy recovery from wind, solar, and hydro is mainly influenced by seasonal, climatic, atmospheric, geographical, and other environmental factors [5]. It is important to note that some of these renewable resources have

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Fig. 1 Organic waste generated from different farming practices. Agriculture framing practices commonly generate straw, stalks, stover, bagasse, shells, husk, hull, and oilseed cakes. Poultry and livestock farming generate waste bedding materials, cattle manure, and poultry litter. Scales, fins, bones, and other trimmings are generated as fish processing wastes

specific environmental, climatic, and geographic restrictions. Therefore, in the quest for sustainable and renewable energy sources, using waste biomass and organic matter to generate bioenergy has emerged as a promising solution to address the global energy crisis and mitigate environmental challenges [6, 7]. Here, we review these different thermochemical and biological waste-to-energy technologies capable of transforming diverse farming waste into solid, liquid, and gaseous biofuels. We review the working principle, benefits and limitations of pelletization, pyrolysis, gasification, liquefaction, anaerobic digestion, and fermentation as viable waste-to-energy methods for utilizing farming waste.

Waste residues from various farming practices

Bioenergy, derived from biological materials like plants and animal waste, is considered a viable alternative to fossil fuels because of its renewable nature and significantly lower greenhouse gas emissions [8]. Among the diverse forms of biomass, agricultural farm residues hold immense potential for bioenergy production due to their abundant availability, low cost, and positive impact on waste management and rural development [9]. The waste materials from various crop farming, crop processing, poultry, and livestock farming have a vast potential to generate multiple biofuels such as bioethanol, biobutanol, biogas, syngas, biomass pellets, and biochar via different conversion techniques. This sustainable approach provides an eco-friendly energy source, contributes to rural development, creates new revenue opportunities, and offers several other socio-economic and environmental benefits as illustrated in Figure 3.

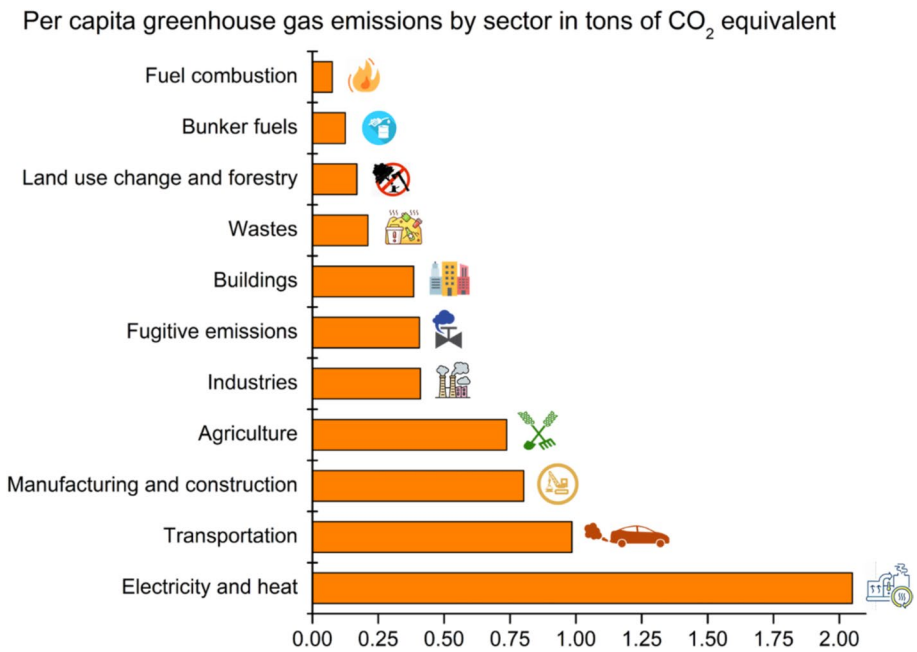


Fig. 2 Current per capita global greenhouse gas emissions by sector. Electricity and heat generation are the most dominating sectors, followed by the transportation sector, in terms of per capita greenhouse gas emissions. Data source: Our World in Data [3]

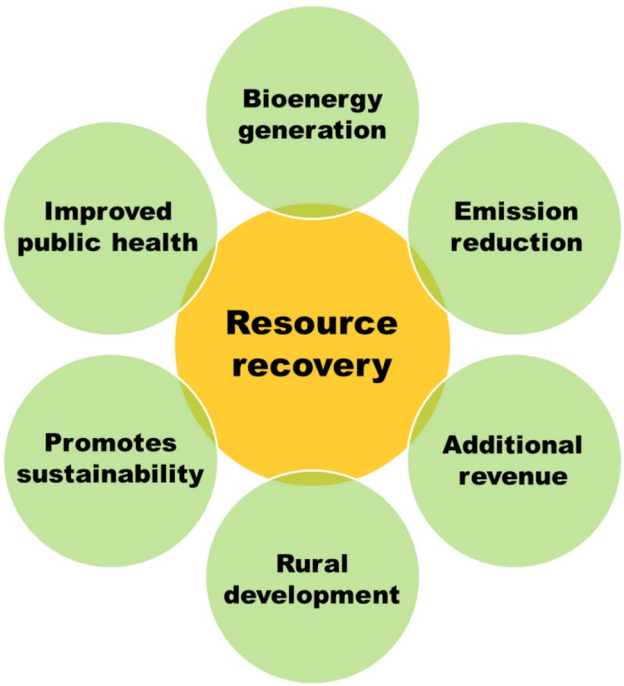


Fig. 3 Socio-economic and environmental benefits of resource recovery from agricultural and farming wastes. Resource recovery from organic waste has significant benefits, such as diversion from landfilling and improper disposal, generating bioenergy, lowering greenhouse gas emissions, creating new revenue streams, rural development, promoting sustainability, and improving public health

Some notable techniques for valorizing waste into energy are pyrolysis, liquefaction, gasification, pelletization, fermentation, and anaerobic digestion. Direct utilization of biomass is restricted because of its physicochemical properties such as high moisture content, low energy value, bulky nature, and difficulty in handling and storage [10, 11]. There are mainly two waste-to-energy conversion methods such as thermochemical and biological. Thermochemical conversion techniques predominantly require heat energy to convert biomass into biofuels and biochemicals. At the same time, the biological methods utilize various microorganisms and enzymes to decompose sugars present in biomass as polysaccharides into small monomers and subsequently to biofuels and biochemicals [12]. The thermochemical conversion process mainly generates liquid fuel as bio-oil and bio-crude oil, solid fuel as biomass pellets and biochar, and gaseous fuels as syngas and hydrogen gas for heat and electricity generation. On the contrary, biological processes generate gaseous biofuels such as biogas, biomethane, biohydrogen, and liquid biofuels such as bioethanol and biobutanol. Biological conversion techniques require more degradation time than thermochemical processes [13]. However, the cost of energy requirement is high in thermochemical methods.

Agriculture

Agricultural residues refer to the byproducts or waste materials generated during agrarian farming. These residues are often left over after the main crop or agricultural product has been harvested or processed. The common agrarian residues are straw, stalks, stover, bagasse, shells, husk, hull, and oilseed cakes. Generally, the straw is used for animal bedding, composting, burning, or as a source of biomass energy. Sugarcane bagasse is a very common crop residue, the fibrous residue left after juice extraction from sugarcane stalks for sugar production, which is commonly used for burning. Alongside the crop residues, there are many processing byproducts such as husks, hulls, and pomace. The outer layer of kernel or cereal crops is known as hull, which accounts for approximately 15–20 wt% of the total grain weight [14]. On the other hand, pomace is the pulpy residue left after extracting juice or oil from fruits such as grapes, olives, and apples. Both husks and pomace can be used as animal feed, bedding, composting, or for bioenergy generation [15]. In addition, various oilseed crops also generate residue during the pressing of oil, known as meal or cake, for example, mustard cake or canola meal. Meal accounts for around 25% of the total crop yield, which contains a small portion of oil, protein, lipid, and fiber [16].

Improper fruit packaging creates trash containing substances including organic acids, essential oils, and phenol, which have antibacterial properties. Fruit residues from grape pomace, apple pomace, and orange peels are exceptionally high in phenol and fructose concentration, making them suitable feedstocks for fermentation to produce biofuels, biochemicals and value-added bioproducts [17–19]. Efficient management and utilization of agricultural residues are crucial to minimize waste, reduce environmental impact, and explore potential value-added applications such as bioenergy production, composting, animal feed, and industrial processes.

Livestock and poultry farming

Livestock and poultry farming require various bedding materials for animal comfort and hygiene. The most used animal bedding materials are straw from cereal crops such as wheat, barley, oats, and rye. Straw provides insulation, absorbs moisture, and offers a comfortable surface for animals to rest on. Wood shavings or wood chips are also used for animal and poultry bedding [15]. They provide good absorption properties, control odor, and offer a cushioned surface. Rice hulls are the outer protective covering of rice grains and can be used as bedding material for livestock, especially poultry. Rice hulls have good absorbency, help control odor, and provide a comfortable surface for animals. Another type of bedding material is pellets, which are produced through

the compression or densification of waste materials such as straw, wood, and paper [20]. They are small, easy to handle, and often used for smaller animals like rabbits, guinea pigs, and poultry. These pellets have several advantages such as absorbency, odor control, and dust reduction.

Biomass pellets used as animal bedding material have high absorbency and can effectively soak moisture and livestock fecal matter. This helps to keep the bedding dry, reducing the chances of bacterial growth and ammonia production. The absorbent nature of pellet bedding helps control odors associated with animal waste. Pellets are compressed material, thus minimizing dust generation, making them suitable for animals with respiratory sensitivities, and contributing to air quality in the animal housing area. Another significant advantage of using pellets as animal bedding is their longevity, which means they last longer than traditional loose bedding materials like straw or wood shavings. The compressed pellets retain their structure and require less frequent replacement.

It is worth mentioning that the choice of bedding material can vary based on the specific livestock, housing conditions, and management practices. Factors such as cost, availability, absorbency, ease of handling, and animal health considerations play a role in selecting the appropriate bedding material for a particular livestock operation [21]. Both livestock and poultry farming generate vast amounts of waste bedding material, meaning the items are discarded after several usages. Those waste materials involve straw, sawdust, wood chips, and pellets that can be further utilized for the generation of biofuel or bioenergy via various conversion techniques.

Other familiar animal waste sources are manure, dung, and litter [22]. Animal manure has become a significant source of surface and groundwater pollution since it is a primary carrier of excreted and undigested nutrients including microorganisms, antibiotics, minerals, hormones and pesticides. Different animal species produce different amounts and types of animal excrement [23]. The precise proportion excreted within a species depends on several variables including feed composition, animal performance, size, age, sex, and husbandry techniques.

The yield and composition of animal manure are highly influenced by animal size and performance, feed conversion, and diet composition. Traditionally, animal manure is either processed and sold as an organic and natural fertilizer in agriculture or used as a raw material to produce biofuels such as dried manure cakes and biogas through anaerobic digestion [24]. However, new waste management and dumping procedures are required by environmental regulations that strictly regulate the pollution of surface and groundwater, nutrient management, odor, and soil contamination. This creates additional encouragement for the utilization of animal manure to produce bioenergy.

Poultry farming includes the cultivation of chickens, ducks, geese, and pigeons. The significant waste from poultry farming includes poultry feathers, manure, litter, poultry processing waste, or poultry slaughterhouse waste. Feather generally accounts for a substantial amount of live weight, which also causes various environmental challenges due to improper disposal strategies. Waste feathers mainly contained keratin protein. Chicken feathers typically contain approximately 91% keratin protein, 8% water, and 1% lipids [25]. Feathers are frequently used to produce feather meal, bedding materials, fertilizers, and decorative items. However, along with other usages, there is an excellent opportunity to convert waste feathers into biofuel. Another significant waste from poultry farming is manure and litter. Manure yields depend on age, the health conditions of the birds, and farm management practices. Various solid waste and byproducts are also produced from slaughterhouses such as blood, skin, heads, feet, gizzards, intestines, and glands. These wastes are commonly employed as recycled feed and as a surface covering for agricultural lands [26].

Pisciculture

Another vital source of waste generation is pisciculture, also known as fish farming or aquaculture waste, which refers to the organic and inorganic byproducts generated from fish farming operations. It primarily includes solid waste, uneaten feed, excrement, and excess nutrients accumulated in the fish farm systems [27]. The proper management of fish farming waste is essential. Fish excreta and uneaten feed release nutrients such as nitrogen and phosphorus into the water. If not properly managed, these excess nutrients can lead to water pollution and eutrophication, causing harmful algal blooms and oxygen depletion. Moreover, aquaculture waste can negatively impact water quality by increasing turbidity, reducing dissolved oxygen levels, and altering pH and nutrient concentrations [28]. Poor water quality can harm fish health, compromise their growth, and increase the risk of diseases. Additionally, high-nutrient fish effluents may cause ecological imbalances and harm aquatic ecosystems by promoting algae growth.

The consumption of fish products has significantly intensified in recent years, resulting in fish being acknowledged as an essential part of a balanced and healthy diet [29]. Hence, there has been a significant expansion in the amount of fish waste generated globally. This rise in fish waste poses substantial economic and environmental concerns throughout the chains of fish distribution. Large quantities of landed fish are dropped or wasted between the landing and consumption stages, commonly known as post-catch fish losses [30]. This issue has become a pressing challenge that requires attention, considering its detrimental impact on the economy and the environment.

The fish processing industry is one of the significant sources of fish waste. According to Hou *et al.*, before being marketed, more than 70% of the fish that are captured go through extra processing steps [31]. This generates a significant amount of fish waste ranging from approximately 20–80% depending upon the level of processing, such as gutting, scaling, and filleting, as well as the fish species. Each fish species has its unique configuration, size, shape, and chemical properties, contributing to the amount of waste produced [32].

The byproducts of fish processing that are generally thrown away may contain viscera, muscle trimmings, heads, skin and fins, scales and bones which account for 12–18%, 15–20%, 9–12%, 1–3%, 5% and 9–15%, respectively [30]. Large fish enterprises use fish processing to improve the stability and quality of their goods while lowering the expense of transporting unusable fish parts. Companies work to maintain product safety by eliminating components like viscera, which could contain germs and enzymes that pose a concern during processing and storage. Therefore, proper management of the waste is essential.

Since fish waste contains nearly the same quantity of proteins as fish meat, it is mainly utilized to produce feed for domesticated animals. Additionally, various fertilizers, including commercial fertilizers, can be manufactured from processed fish waste because of the nutritional quality of fish excrement. Furthermore, fish waste contains much biodegradable organic matter that can be recycled into waste-activated sludge to increase methane generation during anaerobic co-digestion [33].

This enormous waste can generate energy, which can be a good substitute for fossil-based fuel. Fish waste has widely been explored to enhance gas yield via anaerobic digestion. Wu and Song [33] conducted a co-digestion study involving fish waste and waste-activated sludge. They noticed that adding fish waste significantly increased the overall gas yield, especially the biomethane yield. For example, cumulative biomethane production increased from 212 to 684 mL methane per gram of volatile solid fed.

Another study was carried out on the anaerobic digestion of fish waste with hydrochar produced from bamboo via hydrothermal carbonization [34]. The authors noticed that the temperature of hydrothermal carbonization significantly influences gas yield. They obtained the highest biogas yield of 292 L/kg of volatile solids at 200 °C at a 1:2 mixing ratio of fish processing waste and hydrochar. At the same time, the net methane energy of 3,410 kJ/kg was increased by 82% compared to that obtained from the digestion of fish processing waste alone.

Methods to convert farming waste into energy

Recently, increasing demand for recycling and energy conversion has been noticed due to the current state of the economy and environment, there is an increasing demand for recycling and energy conservation. To address this, diverse technologies have been developed to convert waste into bioenergy, which involves turning waste materials into various fuels that can be used to generate electricity or power. Multiple options are available for converting biomass and other farm residue or waste into energy, including two main techniques, such as thermochemical and biological conversion. As discussed earlier, thermochemical conversion involves using heat to decompose the organic components in farm waste materials, while biochemical conversion utilizes microorganisms or enzymes to convert waste into valuable energy [35]. Figure 4 highlights different waste-to-energy technologies and biofuel products from farming waste.

Pelletization

Pelletization is a process of densifying biomass waste to produce pellets that can be used for various applications. Raw biomasses, including agricultural crop residues and municipal solid waste, are bulky in volume and have low density, high moisture absorption capacity, and heterogeneity, which restricts their direct usage for various applications. Therefore, the pelletization or densification process is usually carried out to increase the density by 4–10 times and uniformity and reduce volume. Furthermore, the pellet's standard size and uniform shape minimize transportation costs by enabling compact storage, easier handling, and automatic feeding in large-scale operations [36]. Moreover, particulate emission is also very low compared to raw biomass, which increases environmental safety.

Biomass pellets can be used as fuel directly in combustion or co-firing with coal in power plants for generating heat and power, in household heating and cooking as fuel, in pyrolysis for the generation of bio-oil and biochar, and gasification to produce syngas [37]. The pelletized biomass can also be used as animal feed and livestock bedding material. Pellet usually contains low moisture content ranging from 8 to 11% and a high energy value of up to 23 MJ/kg. All these properties enable them to burn with very high efficiency, comparable to traditional fuel, especially coal [38]. Pelletization is gaining popularity in large markets compared to other fuel processing methods due to its simple operation [39].

The densification process involves a few steps such as raw biomass collection, drying, size reduction,

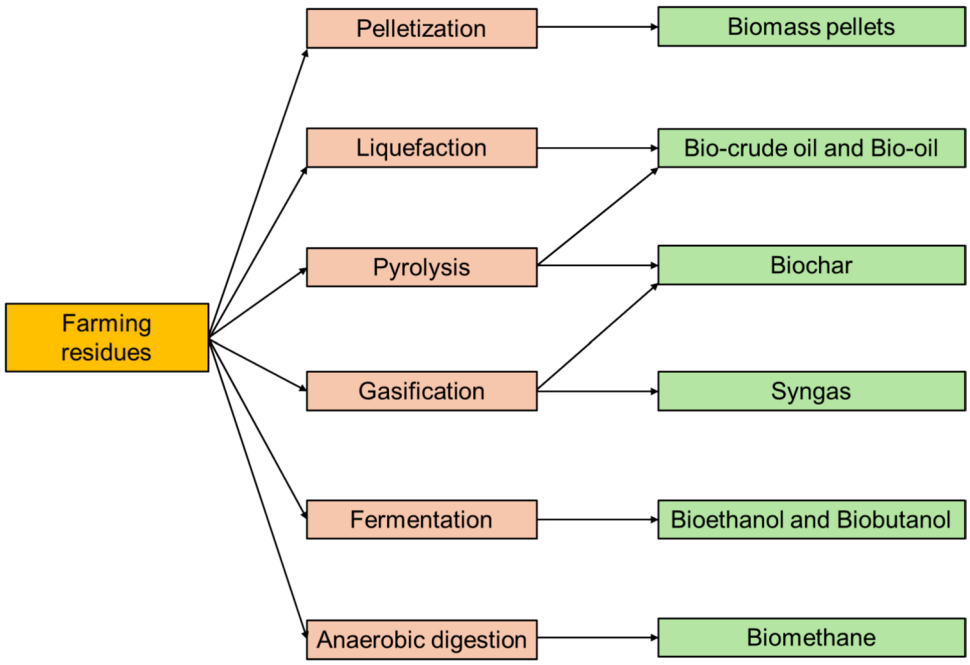


Fig. 4 Biorefinery technologies for the conversion of farming waste into biofuel products. Thermochemical conversion technologies include pyrolysis, gasification, and liquefaction, whereas biological conversion technologies include anaerobic digestion and fermentation. The main products of pelletization, liquefaction, pyrolysis, gasification, fermentation, and anaerobic digestion of farming waste are biomass pellets, bio-crude oil, bio-oil and biochar, syngas, bioethanol and biobutanol, and biomethane, respectively

preprocessing, pretreatment, preparation of pellet formulation, and pelletization [40]. Moreover, biomass's composition and moisture content also affect the pellet quality. Other parameters, such as adding external binders or additives and pelletization process conditions, including pressure, die temperature, die speed, die diameter, and relaxation time, influence fuel pellet quality [41]. Density, durability, and mechanical strength are the basic properties of a pellet that determine its quality. Various bio-based materials such as corn starch, coal tar residue, mustard meal, pyrolysis oil, lignin, lignosulphonate, coffee meal, and amino acid are used as additives for the densification of biomass [39]. Biofuel pellets produced from the densification of the pelletization process use the principles of compression. It involves applying external pressure to the material, which causes the particles to pack more closely together, reducing the void spaces between them. As the temperature and pressure increase, the material undergoes plastic deformation and solid bridge formation, allowing particles to rearrange and fill the gaps, thus resulting in increased density and mechanical strength after cooling [42].

Duca *et al.* demonstrated bio-pellet production from the pelletization of spruce sawdust and olive pruning waste [38]. They used various blends of spruce sawdust with olive pruning and vineyard pruning. They observed that a 20 wt% addition of vineyard pruning increased the durability of spruce sawdust pellets from 78 to 90%, while durability increased up to 92% for olive pruning of 20 wt%. It has been reported that the size of the pellet has a noticeable impact on pellet quality. Valdés *et al.* pelletized coffee husk, rice husk and palm rachis and reported that laminar-shaped and fibrous biomass can produce high yield, higher bulk density, impact resistance, and strong interparticle bonds [43]. However, the water resistance capacities of laminar-shaped biomasses such as rice husk and coffee husk are much higher than that of biomass formed by fibrillary particles.

Papermaking sludge, a byproduct of the paper recycling in paper mills, contains high kaolin and calcite content, which can help produce pellets from very bulky and fibrous material, especially straw [44]. The sludge also contains cellulose fibers, which can positively influence pellets' physical, thermal, and mechanical properties. Matúš *et al.* used papermaking sludge to improve the quality of wheat straw pellets and observed that a 10–20% addition of papermaking sludge can improve the properties of composite pellets [45].

Besides biofuel pellets, the densification process can produce solid fertilizer from various organic waste, especially animal waste. After the solid and liquid separation of animal waste, the solid fraction can be valorized by producing small fertilizer pellets via pelletization. This fertilizer pellet is easy to distribute, requires low transportation costs, and has other environmental benefits. Romano *et al.* demonstrated the production of organic fertilizer pellets from swine manure, swine manure with sawdust, and swine manure with woodchips [46].

Pyrolysis

Pyrolysis is a popular thermochemical conversion process operated under oxygen-deficient environments at temperatures ranging from 350 to 650 °C for several minutes to

hours to decompose the organic material and produce bio-oil, biochar, and gases [47]. The main products of pyrolysis are biochar and bio-oil. However, a trace amount of producer gas is also generated during pyrolysis. The gas generated during pyrolysis comprises CO, CO_2, CH_4, and H_2 with a trace amount of other light hydrocarbons such as ethane, ethylene, propane, and butane [48]. Biochar obtained from pyrolysis can be used as a solid fuel in boilers to produce electricity or heat, catalytic support, precursor for activated carbon, fertilizer, adsorbent for wastewater treatment, generation of several biochemicals as well as raw materials in cosmetics and pharmaceutical industries [49–51].

Pyrolysis can be carried out in various types of biomasses including agricultural crop residues, forestry residues, food waste, municipal solid waste, yard waste, sewage sludge, and animal waste, to name a few. At around 350–550 °C, the organic components in the biomass substrate start to break down during pyrolysis in the presence or absence of air or oxygen to release volatile gases. These volatile gases can be condensed to produce bio-oil and biochemicals. Biomass consists of a long polymeric chain. The pyrolysis process causes the breakdown of larger organic molecules into smaller ones, resulting in gases, condensable liquids such as tar and oil, and solid char released from the process stream. The product yield of pyrolysis and its properties depend on various processing parameters such as temperature, reaction time, type of feedstock, feedstock compositions, heating rate, pressure, and reactor configuration along with design [52].

Pyrolysis can be divided into two major groups, namely fast and slow pyrolysis, based on reaction temperature, heating rate, and vapor residence time. The main products from fast and slow pyrolysis processes are bio-oil and biochar. Fast pyrolysis is carried out at a temperature range of 400–650 °C, a high heating rate of 10–200 °C/s, and a short vapor residence time of less than 2 s [53]. In contrast, slow pyrolysis requires a much longer heating time at a temperature range of 300–500 °C, slow heating rates of 0.1–1 °C/s, and a longer vapor residence time of 10–100 min [54]. Fast pyrolysis can produce 60–75 wt% bio-oil, 15–25 wt% biochar, and 10–20 wt% gas, depending on the composition of the biomass and process conditions. Pyrolysis has a vapor residence time of seconds and milliseconds and can be classified into flash and ultra-rapid pyrolysis. Flash pyrolysis is typically carried out at a temperature range of 700–1,000 °C, with rapid heating rates approaching 700 °C/s and extremely low vapor residence time of less than 0.5 s to produce more than 75% bio-oil [52]. Pyrolysis has been extensively carried out using different agricultural crop residues and other waste streams. Table 1 summarizes a few notable works on the pyrolysis of farming waste to produce biofuels.

Zhang et al. investigated the influences of temperature on the physicochemical properties of biochar obtained from several crop residues including wheat straw, rape straw, corn straw, and rice straw in a tube furnace at 300–600 °C for 60 min under oxygen-deficient environment [64]. The biochar yield decreased with a rise in temperature, with a steady fall over 400 °C. The highest yield obtained for biochar is derived from rice straw because of its high ash content. They noticed that temperature has a significant impact on the properties of biochar, which have a positive relation with carbon content, pH value, electrical conductivity, ash, and surface toughness.

Mukhambet *et al.* conducted slow pyrolysis of flax straw in an electrical tube furnace at 400–600 °C [65]. The highest heating value of flax straw biochar was 18 MJ/kg at 500 °C. Pyrolysis temperature is vital to product yields and quality in various process conditions. In another study, a rise in temperature from 300 to 600 °C increased the electrical conductivity, pH, higher heating value, and fixed carbon of biochar derived from crop residue while decreasing the energy conversion rate [66].

Biswas *et al.* compared the pyrolysis behavior of several agricultural crop residues, including corn cob, wheat straw, rice straw, and rice husk [67]. The maximum oil yield was found for corn cob. After the generation of non-edible fish oil fish waste via transesterification, the de-oiled fish waste was pyrolyzed at temperatures of 350–550 °C for a residence time of 30–150 min in a fixed bed reactor to produce bio-oil and biochar [68]. The optimum pyrolysis conditions of 500 °C temperature, 60 min reaction time, and 0.25 mm particle size resulted in a maximum bio-oil yield of 57 wt%. Another study also performed the pyrolysis of fish waste using a thermogravimetric analyzer and observed that temperature plays a vital role in the decomposition of fish waste [69].

Hassen-Trabelsi *et al.* performed pyrolysis of animal fatty wastes from lamb, poultry, and swine waste in a fixed bed reactor at 400–550 °C at a heating rate of 5, 8, 10, and 15 °C/min to produce bio-oil, biochar and gas [55]. They noticed that the maximum bio-oil yield was obtained at 500 °C with a heating rate of 5 °C/min. The characterization results of the obtained bio-oil show that the physicochemical properties of bio-oil are close to petroleum diesel, which can be directly used as fuel. Besides conventional pyrolysis, the efficiency and yield of pyrolysis can be significantly increased by catalysts. Adding various catalysts during pyrolysis enhances the secondary reactions, reduces the activation energy and energy cost, lowers the reaction time, and affects product distribution, thus increasing the bio-oil yield.

Gasification

Gasification is another promising thermochemical biomass conversion process to generate gaseous fuels such as synthesis gas or syngas. Syngas mainly consists of H_2 and CO, including other gases such as CH_4 and CO_2. Gasification is

Table 1 Studies on the pyrolysis of farming waste into biofuels. These studies demonstrate the effective conversion of waste biomass by pyrolysis to biofuels such as bio-oil, biochar, and gases. Pyrolysis conditions such as temperature, heating rate, and reaction time play vital roles in biomass conversion to biofuels

Feedstock	Operating conditions	Main findings	Reference
Animal fat waste including poultry, lamb and swine waste	• Temperature: 400–550 °C • Reaction time: 20 min • Heating rate: 5–15 °C/min • Reactor: Fixed bed reactor	• Bio-oil produced from animal fat demonstrated satisfactory fuel properties • The highest bio-oil yield of 78 wt% was obtained	[55]
Cattle manure	• Temperature: 650 °C • Heating rate: 10 °C/min • Reactor: Microwave pyrolyzer	• Microwave pyrolysis resulted in greater gas yields from cattle manure than slow pyrolysis • The concentration of CH_4 was highest in contrast to H_2, CO and CO_2 from microwave pyrolysis of cattle manure	[56]
Cattle manure and rape straw	• Temperature: Up to 900 °C • Heating rate: 10–30 °C/min • Reactor: Thermogravimetric analyzer	• Co-pyrolysis of blending feedstock such as cattle manure and rape straw improved the pyrolysis properties when compared to pyrolysis of these feedstocks separately • Lignin decomposition was accompanied by generating aromatic hydrocarbons and their derivatives	[57]
Food waste	• Temperature: 400–550 °C • Reactor: Fluidized bed reactor • Catalyst: Dolomite, red mud and Hydrogen Zeolite Socony Mobil-5 (HZSM-5)	• The addition of catalysts increased the bio-oil yield and fuel properties of bio-oil • The highest bio-oil yield of 49 wt% was obtained	[58]
Pearl millet straw	• Temperature: Up to 900 °C • Heating rate: 10–30 °C/min • Reactor: Thermogravimetric analyzer	• Maximum devolatilization of biomass was found between 220 and 380 °C • Thermal degradation of biomass performance was superior at a 30 °C/min heating rate	[59]
Poultry litter	• Temperature: 460 and 530 °C • Reaction time: 180 min • Reactor: Bubbling fluidized bed	• Ash content in poultry litter reduced the bio-oil yield • The highest bio-oil yield of 28 wt% was obtained	[60]
Rapeseed cake	• Temperature: 400–700 °C • Reaction time: 20 min • Heating rate: 20 °C/min • Reactor: Fixed bed reactor • Particle size: ≤ 0.5, 0.5–1 mm, 1–1.5 mm and 1.5–2 mm	• Bio-oil yield increased with a rise in temperature and a decrease in particle size • The highest bio-oil yield of 35 wt% was obtained	[61]
Sweet potato leaves, stems and peels	• Temperature: 550 °C • Heating rate: 10 °C/min • Reaction time: 60 min • Reactor: Fixed bed reactor	• The use of torrefaction as a pretreatment increased biochar yield while decreasing bio-oil and gas yields compared to pyrolysis alone • Bio-oil obtained from the pyrolysis of leaves had higher nitrogen content	[62]
Swine manure	• Temperature: 400–700 °C • Heating rate: 20 °C/min • Reaction time: 90 min • Reactor: Fixed bed quartz reactor	• Biochar yield was more significant from the pyrolysis of swine manure at 400 °C than at 700 °C • The specific surface area of biochar generated at 500 °C was 37 m^2/g, the highest among all samples • Carbon, hydrogen, nitrogen, and oxygen contents in the biochar samples decreased with increasing pyrolysis temperatures	[63]

usually carried out at higher temperatures, above 800 °C, in the presence of air or inert gases. The gasification process can be classified based on the reaction medium used such as air, steam or water [70]. When the reaction media of the gasification process is air, it is known as air gasification. When the gasification process is carried out in the presence of steam and water, it is termed steam gasification and hydrothermal gasification [71]. Several reactions occur in air

and steam gasification, including pyrolysis, partial oxidation, and steam reforming.

During hydrothermal gasification of biomass, various reactions occur such as water–gas shift, hydrogenation, steam reforming, and Boudouard and methanation reactions [72]. The methanation reaction is responsible for the production of CH_4 where H_2 interacts with either CO or CO_2 [73]. Partial oxidation occurs when there is insufficient oxygen for complete combustion, whereas pyrolysis uses thermal breaking to generate oil, tar, biochar, and gas in an inert atmosphere [74]. In the steam reforming process, CH_4 reacts with steam at higher temperatures and typically in the presence of a catalyst to produce H_2, CO and CO_2 [75]. In the water–gas shift reaction, water and CO combine to form CO_2 and H_2 [76]. CO formation from carbon's reaction with CO_2 defines the endothermic Boudouard reaction [77].

Hydrothermal gasification can be divided into subcritical and supercritical water gasification [78]. When the gasification process uses water below its critical temperature and pressure of 374 °C and 22.1 MPa, it is called subcritical water gasification. Supercritical water gasification involves temperatures and pressures above water's critical points. Supercritical water has excellent solvation capabilities due to its ionic products and free radical mechanism [79]. Hydrothermal gasification is preferred over other thermochemical waste-to-energy processes because of the use of water as a green solvent to convert high moisture-containing biomasses into hydrogen-rich gas fuels. Gasification process parameters such as temperature, reaction duration, feedstock concentration, types of catalysts, catalyst loading, and equivalence ratio have a noticeable influence on gasification performance, gasification efficiencies, lower heating value of gas, the yield of gas, tar and solid, and compositions of gas [80]. Table 2 summarizes some notable studies on the gasification of various farming wastes to produce syngas.

Li *et al.* compared the gasification behavior of several crop residues, including maize straw, rice straw, cotton straw, rice husk, coconut fiber, wood chip, and chicken manure, by conducting steam gasification in a self-designed reactor at 900 °C [90]. Their results show that cotton straw achieved the highest gasification activity. In contrast, rice straw showed the lowest activity, whereas the highest gas yield was obtained for the gasification of rice straw. Taupe *et al.* performed updraft gasification of poultry litter in a fixed bed reactor with an average feed rate of 40 kg/h at 640 °C [91].

Ro *et al.* reported that wet gasification of swine, dairy manure, and sewage sludge can produce a remarkable amount of gas with a high energy content that can be directly used as biofuels [92]. Their results revealed that the highest net energy was found for the wet gasification of swine manure compared to that obtained from other feedstocks. However, all feedstocks provided positive net energy recovery with 90% heat recovery at more than 2% solid content. Hydrothermal gasification is optimal for converting wet biomass into gaseous products [93].

Wang *et al.* supercritical water gasification of corn stalk at 500–800 °C with feedstock concentration of 1–9% for 1–15 min of residence time [94]. According to the findings, boosting the reaction performance of biomass requires a higher reaction temperature, while the reaction duration should be long, and the feed concentration is low. A mechanism investigation revealed that the biomass first undergoes a depolymerization reaction and produces monomers and their derivatives, which are then further cracked and polycondensed into a nucleus to produce carbon microspheres as biomass concentration reaches a critical concentration. Carbon microspheres were reduced as the process progressed due to secondary reactions, coke combustion, and reduction reactions.

Wang *et al.* performed supercritical water gasification of wheat straw and reported that higher temperatures enhanced hydrogen yield with the maximum yield reaching 13 mol/kg at 700 °C for 30 min [95]. Khorasani *et al.* performed supercritical water gasification of dairy wastewater and reported that higher temperatures and longer reaction times enhanced gas yield and gasification efficiencies [96]. They found the optimum gasification conditions for dairy wastewater to be 400 °C and 60 min, with a maximum gas yield of 1.4 mmol/g. Adding manganese oxide and manganese dioxide catalyzed hydrogen generation. However, the highest hydrogen yield of 35% was obtained using manganese dioxide. Moreover, the concentration of formic acid, catalyst addition, and the formic acid concentrations also increased gasification efficiency.

Katsaros *et al.* conducted co-gasification of poultry litter with beech wood at a 50/50 ratio and investigated the impact of equivalence ratio and feedstock type on tar formation [88]. They obtained the highest gas yield for beech wood compared to poultry litter alone and the blend. The highest tar is obtained from beech wood because of its high lignin content. However, due to the high nitrogen content in poultry litter, a noticeable amount of nitrogen-containing tar was found for poultry litter and the blend gasification.

A significant problem with the gasification of agricultural farms and processing residue is the generation of high tar content. The tar generated in the gasification process can be categorized into three groups, namely primary, secondary, and tertiary, depending on how they are formed under varying process conditions [97]. Primary tar, which emerges during pyrolysis at 400–600 °C, comprises primarily oxygenated compounds like levoglucosan, glycolaldehyde, and furfural [98]. With increasing temperature, primary tar components break down and transform into secondary tar, predominantly composed of phenolics and olefins at 700–850 °C [99]. Tertiary tar, produced at even greater

Table 2 Studies on the gasification of farming waste into biofuels. These studies demonstrate the effective conversion of waste biomass by gasification to biofuels such as syngas and biochar. Gasification conditions such as temperature, feedstock concentration, reaction time, equivalence ratio, and catalysts play vital roles in biomass conversion to biofuels

Feedstock	Operating conditions	Main findings	Reference
Barley straw, canola hull and oat hull pellets	• Temperature: 300–500 °C • Biomass-to-water ratio: 1:05–1:15 • Reaction time: 20–60 min • Reactor: Fixed bed reactor	• Total gas yield improved from 1.3 mmol/g to 7.3 mmol/g with the increase in temperature from 300 °C to 500 °C at a biomass-to-water ratio of 1:10 and reaction time of 60 min • Higher temperatures led to thermal cracking, hydrocarbon reforming, and secondary tar cracking reactions with superior syngas yields	[81]
Cattle manure	• Temperature: 700–875 °C • Heating rate: 50 °C/min • Reaction time: 30 min	• The type of process gas, whether oxidative or inert, impacted syngas composition • While temperature variation alone did not show clear trends, its role as a secondary factor with other variables impacted the process performance and syngas yields	[82]
Chicken manure	• Temperature: 550–750 °C • Slurry concentration: 35–65 wt% • Pressure: 23–33 MPa	• An increase in gasification temperature and water-to-biomass ratio, along with lowering pressures, enhanced ammonia production • A decrease in gasification temperature and pressure, along with increasing slurry concentration and the water-to-biomass ratio, boosted the heat supply	[83]
Corn cob	• Temperature: 800–950 °C • Catalysts: Potassium hydroxide, sodium hydroxide, potassium carbonate and sodium carbonate • Reactor: Fixed bed reactor	• Catalysts dramatically increased gas yields • Higher gasification temperature positively impacted gas yields • Potassium hydroxide performed the best among all catalysts	[84]
Food waste	• Temperature: 550–700 °C • Reaction time: 0–30 min • Feedstock concentration: 5–9 wt% • Catalysts: Potassium carbonate, sodium carbonate and Raney-Nickel • Reactor: Autoclave	• The optimum concentration of food waste was 7 wt% • Temperature and residence time positively influence gas yield, carbon conversion and gasification efficiency • The catalysts dramatically increased the conversion of liquid intermediates to gaseous products	[85]
Fruit pulp	• Temperature: 400–600 °C • Reaction time: 40–200 min • Biomass-to-water ratio: 2.5–10 wt% • Catalyst: Potassium hydroxide • Reactor: Batch reactor	• Temperature had the highest impact on gas yield • Gas yields increased with the increment in temperature • Syngas comprised of a mixture of H_2, CO, CO_2 and CH_4	[86]
Horse manure	• Temperature: 400–600 °C • Reaction time: 15–45 min • Biomass-to-water ratio: 1:5, 1:10 • Pressure: 23–25 MPa • Reactor: Tubular batch reactor	• Optimum gasification conditions are 600 °C, 45 min with 1:10 biomass-to-water ratio • A decrease in feed concentration enhanced the gas yield due to improved water–gas shift reaction at high temperatures	[22]
Livestock manure, food wastewater, digested sludge	• Temperature: 600–1,000 °C • Heating rate: 26–31 °C/min • Reaction time: 30 min	• Livestock manure resulted in a gas yield of 44 wt% at 1,000 °C • The carbon conversion efficiency and carbon gasification efficiency improved with higher gas yields and lower tar and biochar generation	[87]
Poultry litter	• Temperature: 750 °C • Equivalence ratio: 0.16–0.28 • Reactor: Bubbling fluidized bed reactor	• A rise in the equivalence ratio increased the heating value of gas • The results indicated the potential of poultry litter as a suitable feedstock for syngas production	[88]
Swine manure	• Temperature: 580–660 °C • Reaction time: 1–45 min	• The study investigated the directional transformation of nitrogen in pig manure • The conversion and stability of nitrogen predominantly relied on catalysts rather than gasification temperature	[89]

temperatures of 850–1,000 °C, contains complex polyaromatic hydrocarbons, which are exceptionally stable and possess a high dew point [100]. Various catalysts such as dolomite, olivine, biochar, nickel, iron, and alkali oxides are employed in biomass gasification to reduce the tar content [101, 102].

Liquefaction

Liquefaction is an emerging method for transforming various waste feedstocks into liquid fuels at moderate temperatures of 200–370 °C under a high-pressure range of 4–20 MPa [103]. In the liquefaction process, numerous solvents are usually used to soften the organic matter via processes like solvolysis, hydrolysis, and cracking to produce bio-crude oil. Temperature, reaction time, feedstock/water ratio, and catalysts are primary factors affecting the reaction mechanisms, performance, and product yields in liquefaction. Hydrothermal liquefaction is suitable for the conversion of biomass containing high moisture content. Bio-crude oil produced from liquefaction possesses high heating value, low oxygen content, improved thermal stability, high carbon content, more excellent flowability, low acidity, and low polymerization, which makes it superior to the bio-oil obtained from the pyrolysis process [104]. Bio-crude oil has several applications such as directly as a liquid biofuel or as a feedstock to synthesize green diesel and jet fuels. In hydrothermal liquefaction, hydrolysis of biomass or other organic matter occurs to generate smaller monomers. This is followed by the cleavage and decarboxylation of smaller monomers into intermediate volatile compounds, which finally condense and polymerize to generate green hydrocarbons.

In hydrothermal liquefaction, water is used as a green solvent due to its availability, low cost and environmentally friendly nature. However, ethanol, methanol, acetone, and benzene have also been employed as co-solvents in hydrothermal liquefaction in addition to water [105]. Different solvent types have varying densities and extraction capacities. Therefore, choosing and using the proper solvent for hydrothermal liquefaction biomass conversion can considerably increase the yield of bio-oil and the generation of fuel and chemicals. Types of solvents and catalysts influence the liquefaction process of biomass and other organic materials by reducing the severity of the liquefaction process and enhancing heat and mass transfer. Water, toluene, acetone, and alcohol, such as methanol, ethanol, and propanol, are the most used organic solvents because of their low boiling points and ease of separation. Liquefaction results in a primary liquid product containing tar and the watery phase. The tar phase mainly contains bio-crude oil, while the watery phases contain water and various biochemicals, which can be recycled, reused, or recovered for other value-added applications [106]. Table 3 summarizes some notable studies on the hydrothermal liquefaction of farming wastes to produce bio-crude oil.

Animal byproducts are a promising feedstock for the hydrothermal liquefaction process for the generation of bio-crude oil. Conti *et al.* conducted hydrothermal liquefaction of swine manure, sewage sludge, dairy cow manure, and fish sludge in subcritical and supercritical conditions with and without catalysts [117]. The highest bio-crude oil yield of 50 wt% was obtained from fish waste, followed by sewage sludge, which resulted in a 45 wt% bio-crude oil yield. León *et al.* studied hydrothermal liquefaction of animal byproducts such as skin, fat, tendons and bones from porcine and bovine in an electrically heated lab at 150–290 °C for 5–15 min residence times under high pressure of 1–2.8 MPa [107]. The highest yield of bio-crude oil was attained at a temperature of 225 °C, while the higher heating value of bio-crude oil varied in the range of 35–39 MJ/kg. Higher temperature enhances free fatty acid yield along with amides and heterocyclic compounds, while lower temperature generates triglycerides and fatty acids.

Bayat *et al.* explored the influence of various process parameters of hydrothermal liquefaction on product yields of food waste where the temperature varied from 240 to 295 °C for 0–60 min of reaction time [114]. Hydrothermal liquefaction temperature significantly influences the yield and properties of bio-crude oil from fruit peel. Cheng *et al.* reported a relatively low yield of bio-crude oil at the highest temperature while conducting hydrothermal liquefaction at 300–350 °C for a residence time of 30–120 min [118]. Motavaf and savage studied isothermal and fast hydrothermal liquefaction processes at 200–600 °C [119]. They reported a higher yield of bio-crude oil from fast hydrothermal liquefaction. Hongthong *et al.* examined the co-hydrothermal liquefaction of pistachio hulls and polypropylene [120]. Their study demonstrated that hydrothermal liquefaction may be an effective method for valuing plastic and food waste. The process efficiency was significantly increased by adding hydrogen donor solvents and catalysts at standard temperatures of about 350 °C.

The type of solvent is also a vital parameter in the case of hydrothermal liquefaction of organic biomass. Biswas *et al.* investigated the impact of various solvents, including water, ethanol, and methanol, on the product yields from the hydrothermal liquefaction of corncobs [111]. The authors experimented with three different temperatures, such as 260, 280 and 300 °C, while the reaction holding duration remained the same at 15 min for all cases. For the synthesis of bio-oil, the measured efficiency of the solvent used in the liquefaction of corn cob followed the decreasing order of water, methanol, and ethanol. The maximum level of phenolic compounds was obtained when the solvent was water (60%) followed by methanol (57%) and ethanol (56%).

Table 3 Studies on the liquefaction of farming waste into biofuels. These studies demonstrate the effective conversion of waste biomass by liquefaction to bio-crude oil. Liquefaction conditions such as temperature, reaction time, and solvents play vital roles in biomass conversion to bio-crude oil

Feedstock	Operating conditions	Main findings	Reference
Animal byproducts including skin, fat, tendons, and bones	• Temperature: 150–290 °C • Reaction time: 5–15 min	• The composition of bio-crude oil is significantly impacted by the process temperature as free fatty acids and amides rise and triglycerides fall with temperature • The highest yield of bio-crude oil was 61 wt%	[107]
Animal manure	• Temperature: 310–340 °C • Reaction time: 30 min	• The highest bio-crude obtained for swine manure • A significant amount of heavy metal is produced • The highest yields of bio-crude oil from sheep manure, swine manure, beef manure, cattle manure, broiler manure and laying hen manure were 26 wt%, 31 wt%, 20 wt%, 19 wt%, 18 wt%, and 21 wt%, respectively	[108]
Barley, canola, flax and wheat straw	• Temperature: 270–300 °C • Reaction time: 15–45 min	• Barley straw resulted in higher bio-crude oil yield in the presence of potassium carbonate as the catalyst • Nearly 43 wt% of bio-crude oil was obtained at 270 °C and a reaction time of 15 min with 5 wt% of catalyst loading when the biomass-to-water ratio was 1:10	[109]
Cassava rhizome	• Temperature: 275 °C • Reaction time: 15 min	• Recirculation of alkaline catalysts such as potassium carbonate and sodium carbonate and water-soluble products improved the yield and quality of bio-crude oil • Acetic acid was the primary compound in water-soluble products from hydrothermal liquefaction	[110]
Corn cob	• Temperature: 260–300 °C • Reaction time: 15 min	• The highest yield was obtained when the solvent was water, followed by methanol and ethanol • The highest yield of bio-crude oil was 38 wt%	[111]
Corn stalk	• Temperature: 170 °C • Reaction time: 0–240 min	• Bio-crude oil contained essential hydrocarbons, acids, alcohols, esters, ketones, and ethers • Unstable free radicals generated by cellulose degradation reactions were observed	[112]
Food waste	• Temperature: 280–340 °C • Reaction time: 30 min	• Bio-oil yield and energy recovery are greatly influenced by temperature and mixing ratio • The highest yield of bio-crude oil was 40 wt%	[113]
Food waste	• Temperature: 240–295 °C • Reaction time: 0–60 min	• Higher temperatures decreased the yields of bio-crude oil containing greater polycyclic, oxygen and nitrogen-containing aromatic compounds and higher ignition temperatures • The highest yield of bio-crude oil was 28 wt%	[114]
Poultry manure	• Temperature: 200–350 °C • Reaction time: 30 min	• Acid-catalyzed hydrothermal liquefaction of poultry manure at 300 °C in 30 min produced bio-crude oil with improved heating value as a liquid fuel • Synergistic effects were observed that led to improved bio-crude quality and increased oxidation of the organic components in the aqueous phase under an acidic environment	[115]

Table 3 (continued)

Feedstock	Operating conditions	Main findings	Reference
Spent mushroom substrate and tomato residues	• Temperature: 300 °C • Reaction time: 90 min	• Liquid product obtained from hydrothermal liquefaction of feedstocks demonstrated effective anti-fungal properties • The liquid product can be used as an anti-fungal agent against wood-decaying fungi with applications in wood preservative formulations	[116]

Anaerobic digestion

Anaerobic digestion, or biomethanation, is a simple and mature bioconversion process of biomass to biogas. In this process, organic solids undergo microbial decomposition and stabilization under anaerobic conditions to produce biomethane. The involvement of various groups of anaerobic microorganisms transforms organic matter into biogas, a renewable energy source for generating heat, electricity, or power [121]. Recently, there has been a growing interest in improving and innovating the anaerobic digestion process to co-produce biomethane and biohydrogen as high-value gaseous biofuels [122].

Biogas is a combustible fuel gas product from anaerobic digestion that can be used for cooking, power generation, and heating homes. The most common use of biogas is for cooking and lighting households. Biomethane possesses a calorific value close to 36 MJ/m^3, which can quickly be burned to release heat energy. Digestate, liquid effluent, or bio-slurry is the byproduct of anaerobic digestion, which is nutrient-rich [123]. Hence, it can be used as fertilizer for crop production. Digestate is very high in nutrients, especially nitrogen, potassium, and phosphorus, which are the essential components of fertilizer. Besides fertilizer for crop farming, it can be used as a feedstock for hydrothermal liquefaction or hydrothermal gasification, considering its high water content, for generating bio-crude oil and syngas.

Anaerobic digestion typically involves biomass hydrolysis, acidogenesis, acetogenesis, and methanogenesis to produce biogas [124]. Firstly, high molecular weight organic substrates such as carbohydrates, lipids, and proteins are degraded into monomeric units with water solubility and low molecular weight compounds like sugars, fatty acids, amino acids, and glycerol. Several bacterial species are involved in a biogas process when polysaccharides are hydrolyzed. These microorganisms contain many enzymes integrated into cellulosomes in the microorganisms' cell walls. The pH, temperature, composition, substrate particle size, and intermediate product concentrations impact the rate of hydrolysis. Hence, enzymes are frequently added to speed up the hydrolysis of these carbohydrates.

During the acidogenesis phase, the fermentative microorganism uses the sugars, fatty acids, and amino acids from hydrolysis to make volatile organic acids such as acetic, propionic, butyric, fatty acids, alcohol, H_2, and CO_2 [125]. The trace amount of byproducts generated include NH_3, CO_2 and H_2S. Almost all the microorganisms participating in the hydrolysis stage also participate in fermentation. Acetogenesis is the process by which volatile organic acids are converted to acetate, CO_2, and H_2 in the acid-forming stage. Numerous acids and alcohols, including butyrate, propionate, and ethanol, can be converted into acetate as a substrate by methanogenic bacteria [126].

The most crucial stage of the anaerobic digestion process is methanogenesis, where methanogenic bacteria consume available intermediates such as CO_2 and H_2 to generate CH_4 and CO_2. Acids, alcohols and other nitrogenous compounds cannot be used directly by methanogenic bacteria. Hence, these components accumulate in the digester supernatant. Approximately 70% of biomethane is produced from acetate by acetoclastic methanogens and 30% of H_2 and CO_2 by hydrogenotrophic methanogens [127]. These obligate anaerobic bacteria are susceptible to environmental changes, while hydrogen-utilizing methanogens are more tolerant.

The final fermentation products in the acidogenesis phase influence the methanogenesis phase's loading, efficiency and operating stability. Methane is synthesized in the last step of methanogenesis by two kinds of methanogenic bacteria. The first group breaks acetate into CH_4 and CO_2, whereas the second group produces CH_4 by using H_2 as an electron donor and CO_2 as an acceptor. Consequently, the three primary production routes for biomethane production are [128]. Methane can be produced through the decarboxylation of methyl alcohols, methyl amines, and methyl sulfides. In acetoclastic or acetotrophic methanogenesis, biomethane is produced by acetate decarboxylation [129]. Numerous factors affect the product yields and anaerobic digestion process, including biological oxygen demand, chemical oxygen demand, carbon/nitrogen ratio, volatile solids, hydraulic retention time, loading rate, temperature, pH value, mixing, organic loading, and inoculum [130]. Pretreatment of biomass samples before digestion also affects the digestion process and gas compositions. Temperature plays an evident role in the anaerobic digestion process, and the temperature regimes of anaerobic digestion can be classified into mesophilic and thermophilic.

Mesophilic digestion runs at a moderate temperature of 35 °C where the digestion rate is slow, and gas yield is low. Still, it remains alluring due to their lower heating energy cost than thermophilic digesters [131]. On the contrary, thermophilic digestion requires a comparatively higher temperature of 55 °C, which enhances the reaction rate and thus increases the gas production [132]. The anaerobic digestion of biomass wastes can be carried out as mono-digestion or co-digestion [133]. The process of anaerobic co-digestion offers several benefits over mono-digestion such as enhancing digestion and energy generation. This is achieved by providing more nutrients to the microorganisms and decreasing inhibitory chemical toxicity by diluting co-substrates [134]. Mono-digestion is commonly used for smaller biogas production facilities that digest animal manure. In co-digestion, several feed materials are digested simultaneously in the same reactor. The co-digestion process is now well established due to its thermal stability and high efficiency. Co-digestion of various substrates has significantly improved biogas production potential compared to single substrates alone [134].

Aravani *et al.* conducted co-digestion of waste watermelon, cattle manure, corn silage, and tomato processing residues to produce biogas [135]. At the same time, anaerobic digestion was carried out in mesophilic conditions in a continuously stirred tank reactor for a retention time of 20 days. Biomethane generation increased with an increase in total volatile solid content. Table 4 summarizes some notable studies on the anaerobic digestion of various farming wastes to produce biogas.

Fermentation

Fermentation is a metabolic process in various microorganisms, such as yeast and bacteria, that produces energy without oxygen or in anaerobic conditions. During fermentation, microorganisms break down sugars or other organic compounds into simpler products, usually alcohol, acids, or gases, releasing energy as adenosine triphosphate. Fermentation of biomass for ethanol production is a widely used process to produce bioethanol, a renewable and sustainable fuel alternative to gasoline. Sugar-containing substrates can be converted during fermentation into various biofuels and biochemicals, depending on the fermenting microorganism and its metabolic bioprocesses [143]. During fermentation, microorganisms utilize sugars to produce multiple end products such as acetic acid, butyric acid, lactic acid, bioethanol, biobutanol, and other bioproducts [144]. After hydrolysis, the resulting hydrolysate mainly contains glucose and xylose, essential raw materials for fermentation to produce bioethanol, biobutanol, and other biochemicals.

The bioconversion process to produce bioethanol involves several steps, including saccharification or hydrolysis, fermentation, and distillation [145]. Saccharification or hydrolysis is the breaking down complex carbohydrates into simple sugars using enzymes, making it easier for yeast to ferment them. The sugars obtained from biomass hydrolysis are used as the carbon source by the fermenting microorganisms to produce bioethanol, biobutanol, and other bioproducts. Several bacteria and yeasts can efficiently ferment these sugars to produce ethanol, using them for their growth and reproduction. Microorganisms that successfully ferment xylose under laboratory conditions often yield lower results when working with lignocellulosic hydrolysates. *Saccharomyces cerevisiae* is the most used yeast species for bioethanol production due to its robustness and proficiency in fermenting glucose derived from lignocellulosic biomass hydrolysis [146]. Another widely used microorganism for industrial-scale bioethanol production is *Zymomonas mobilis*, which exhibits higher yield and productivity but is limited in usage due to a higher risk of contamination [147].

The fermentation process is influenced by temperature, retention time, pH, yeast strain, and the feedstock's composition and biomass pretreatment [148]. Based on how hydrolysis and fermentation are combined, many configuration approaches include separate hydrolysis and fermentation, pre-saccharification and simultaneous fermentation, and saccharification and simultaneous fermentation [149]. High glucose concentrations have the disadvantage of preventing enzyme action. Other limitations of this process are high cost and generation of inhibitors from cellulase and glucose. Saccharification and simultaneous fermentation can solve the problems associated with separate hydrolysis and fermentation as sugars are fermented as soon as they are produced, thus reducing the chances for inhibitor generation [150]. The other benefits of this process are low investment costs, less time requirements, and low risk of contamination. The main disadvantage of the saccharification and simultaneous fermentation procedure is that the enzymatic yield is not maximized because the temperature utilized in this process is lower than the temperature at which the enzymes act best, about 35 °C, to prevent the destruction of yeast. When hydrolysis and fermentation of sugars co-occur in the same reactor, it is called co-fermentation.

Agricultural crop and farm residues, along with corn starch, sugarcane, and high starch containing materials, are widely used for ethanol production. Tran *et al.* produced ethanol and polyhydroxybutyrate from the fermentation of various lignocellulosic biomass such as sugarcane bagasse, silver glass, and cardboard boxes using *S. cerevisiae* [150]. Agricultural residues have the potential to produce around 5.5 billion liters of bioethanol by introducing second-generation bioethanol plants near processing industries [151]. Lifecycle and techno-economic analyses are essential to determine the technical and economic feasibility, carbon emissions, resource recovery, residue management, and

Table 4 Studies on the anaerobic digestion of farming waste into biofuels. These studies demonstrate the effective conversion of waste biomass by anaerobic digestion to biogas. Anaerobic digestion conditions such as temperature, hydraulic retention time, and feedstock concentration play vital roles in biomass conversion to biogas

Feedstock	Operating conditions	Main findings	Reference
Chicken manure	• Temperature: 37 °C • Hydraulic retention time: 21 days • Reactor: Semi-continuous reactor	• Biochar addition to anaerobic digestion enhanced CH_4 production by 8.7-fold due to maintaining a higher chemical oxygen demand removal rate and total ammonia nitrogen concentrations below inhibitory levels • Propidium monoazide uncovered biochar's role in supporting the viability of methanogenic microorganisms	[136]
Corn silage and cow manure	• Hydraulic retention time: 13.3 days • Organic loading rate: 8.04–8.3 kg volatile solid/m^3/day • Reactor: Semi-continuous pilot scale reactor	• Pretreatment significantly enhanced the biogas generation • The highest yield of biogas was 0.236 L/g of volatile solids	[124]
Cow manure	• Temperature: 20 and 28 °C • Hydraulic retention time: 28–30 days • Reactor: Batch reactor	• CH_4 yield increased by 50% when temperature rose from 20 °C to 28 °C • Around 90% of pathogens were removed • The highest biogas yield was 0.23–0.29 L/g of volatile solids	[132]
Cow manure	• Temperature: 37 °C • Solid retention time: 30 days • Reactor: Batch reactor	• CH_4 yields increased to 92% when the organic loading was 40 L/g of volatile solids • The increased duration observed under high organic loading conditions indicated a rise in the quantity of biodegradable substrates	[137]
Cow manure	• Temperature: 36 °C • Solid retention time: 135 days • Reactor: Semi-continuous reactor	• This study demonstrated that combining anaerobic digestion with vacuum stripping and absorption enhanced anaerobic digestion and biogas yields • Post-treatment using vacuum stripping also enhanced the recycling of manure digestate by reducing the occurrence of pathogens	[138]
Fish waste and activated sludge	• Temperature: 37 °C • Hydraulic retention time: 0–50 days • Feed concentration: 0–10 vol.% • Reactor: Batch reactor	• The addition of fish waste remarkably augmented the CH_4 generation • The highest yield of biogas was 684 mL/g volatile solids	[33]
Food waste	• Temperature: 35–55 °C • Hydraulic retention time: 5–9 days • Reactor: Two-stage continuously stirred reactor	• A recirculation rate of 0.3 reduced the alkali requirement by 54% • The highest productivity of biogas was 2.9 L/day	[139]
Rice straw	• Temperature: 35 °C • Organic loading: 9.1 g/L volatile solid • Reaction time: 22 days	• Briquetting of rice straw as a pretreatment process improved biogas production • When briquetting was combined with sodium hydroxide treatment, CH_4 yield increased by 26% compared to the base case	[140]
Swine manure, algae and activated sludge	• Temperature: 20–33.4 °C • Reaction time: 28 days	• The gradual scaling-up process involving anaerobically digested swine manure using *Chlorella vulgaris* and bacteria in activated sludge improved anaerobic digestion performance • The integrated treatment process demonstrated effective potentials for treating swine manure in a scale-up system to produce biogas	[141]

Table 4 (continued)

Feedstock	Operating conditions	Main findings	Reference
Swine manure and crude glycerol	• Inoculum-to-substrate ratio: 1:1, 2:1 and 4:1 • Swine manure-to-crude glycerol ratio: 99:1 and 97:3 wt% • Hydraulic retention time: 17 and 22 days • Reactor: Continuously stirred reactor	• Biogas productivity increased by 222% for a 3% addition of crude glycerol • The highest yield of biogas was 0.6 L/g of volatile solids	[142]

marketability of products obtained from farming waste regardless of thermochemical, hydrothermal, or biological conversion technologies [152].

Conclusion

Utilizing farm waste as feedstock for alternative fuels or bioenergy can decrease reliance on petroleum-based fuels, lower waste management costs, and create employment opportunities. The primary methods for converting biogenic residues into usable biofuels are thermochemical processes, including pyrolysis, liquefaction, and gasification, as well as biological processes, especially anaerobic digestion and fermentation. Thermochemical methods are generally more efficient due to faster operations and a lower risk of toxic byproducts. Key factors influencing conversion efficiency include temperature, reaction time, feedstock concentration, and reactor type, while the chemical composition of feedstock is crucial for biological conversion. Pyrolysis mainly produces bio-oil, liquefaction produces bio-crude, and gasification generates syngas. To secure a sustainable and affordable bioenergy supply, it is essential to develop cost-effective methods, minimize byproducts, improve technology to reduce energy consumption, and study biomass characteristics. This review highlights current practices, prospects, and challenges in managing biogenic residues from agriculture, livestock, poultry, and fish farming. With the decreasing fossil fuel supply and rising demand for sustainable energy, the future of biofuels in the fuel industry looks promising.

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Declarations

Conflict of interest Sonil Nanda declares that he is an Editor of Environmental Chemistry Letters. The authors have no conflict of interest to disclose.

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